

Unit 6, Lesson 3: Determining Proportional Relationships

Benchmark

MA.B.AR.3.1 Determine if a linear relationship is also a proportional relationship.

Learning Target

- ☐ I can determine if a linear relationship is also a proportional relationship given a graph, table, equation, or written description.

6.3.1 Warm-Up: Would You Rather?

Consider the following situations:

Situation A:

The cost to make 36 cupcakes is \$10. You charge \$0.50 per cupcake.

Situation B:

The cost to make 36 cupcakes is \$8.00. You charge \$18.00 for all three dozen cupcakes.



- Which situation would you rather be in?
- Explain or show your work to justify your reasoning.

6.3.2 Exploration: Exploring Proportional vs. Non-Proportional Relationships

Work with your partner to complete the following.

- Diego earns \$10 per hour babysitting. He has no money saved before he starts babysitting and plans to save all of his earnings. Complete the table with Diego’s savings.

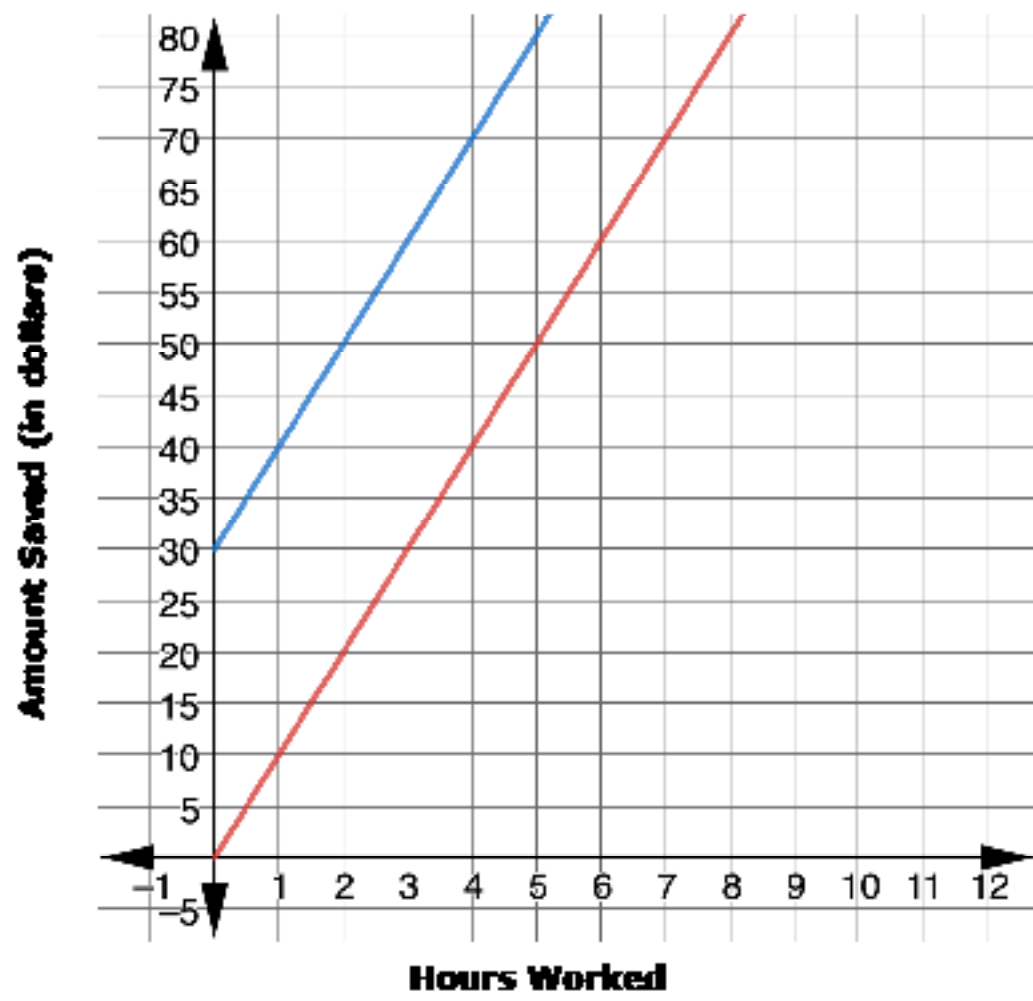
Hours Worked	Amount Saved
0	0
1	
2	
4	
6.5	
11	
x	

- Diego’s sister, Camilla, started with \$30 saved before she starts babysitting. She earns \$10 per hour as well and saves all that she earns. Complete the table with Camilla’s savings.

Hours Worked	Amount Saved
0	30
1	
2	
4	
6.5	
11	
x	

Diego’s savings are modeled by the **red line**.

Camilla’s savings are modeled by the **blue line**.



3. Using evidence from the tables and the graph, explain which line represents a proportional relationship.

4. Graphs of proportional relationships meet two criteria. Complete the statements to describe both criteria.
- a. The graph must be _____.
 - b. The graph must cross the coordinates (____, ____).

The equations used to model the savings for Diego and Camilla are shown.

Diego’s Savings	Camilla’s Savings
$y = 10x$	$y = 10x + 30$

5. Discuss with your partner the difference between the two equations and explain how to determine if an equation represents a proportional relationship or not.

6.3.3 Guided Practice: Determining if a Relationship is Proportional

1. Identify if each equation is proportional or not.

If it is proportional, circle the constant of proportionality in the equation. If it is not proportional, box the portion of the equation that indicates the relationship is not proportional.

Equation	Proportional or Not Proportional?
$y = 8 + x$	☐ proportional ☐ not proportional
$y = -4x$	☐ proportional ☐ not proportional
$y = 5 + \frac{1}{2}x$	☐ proportional ☐ not proportional
$y = \frac{x}{5}$	☐ proportional ☐ not proportional
$y = 6 - 2x$	☐ proportional ☐ not proportional
$y = -0.75x$	☐ proportional ☐ not proportional
$y = \frac{3}{x}$	☐ proportional ☐ not proportional

2. Jada's grandparents started a savings account for her in 2010. The table shows the amount in the account each year.

Years Since 2010	Amount (in dollars)
0	600
2	750
4	900
6	1050

Explain whether this is a proportional relationship or not.

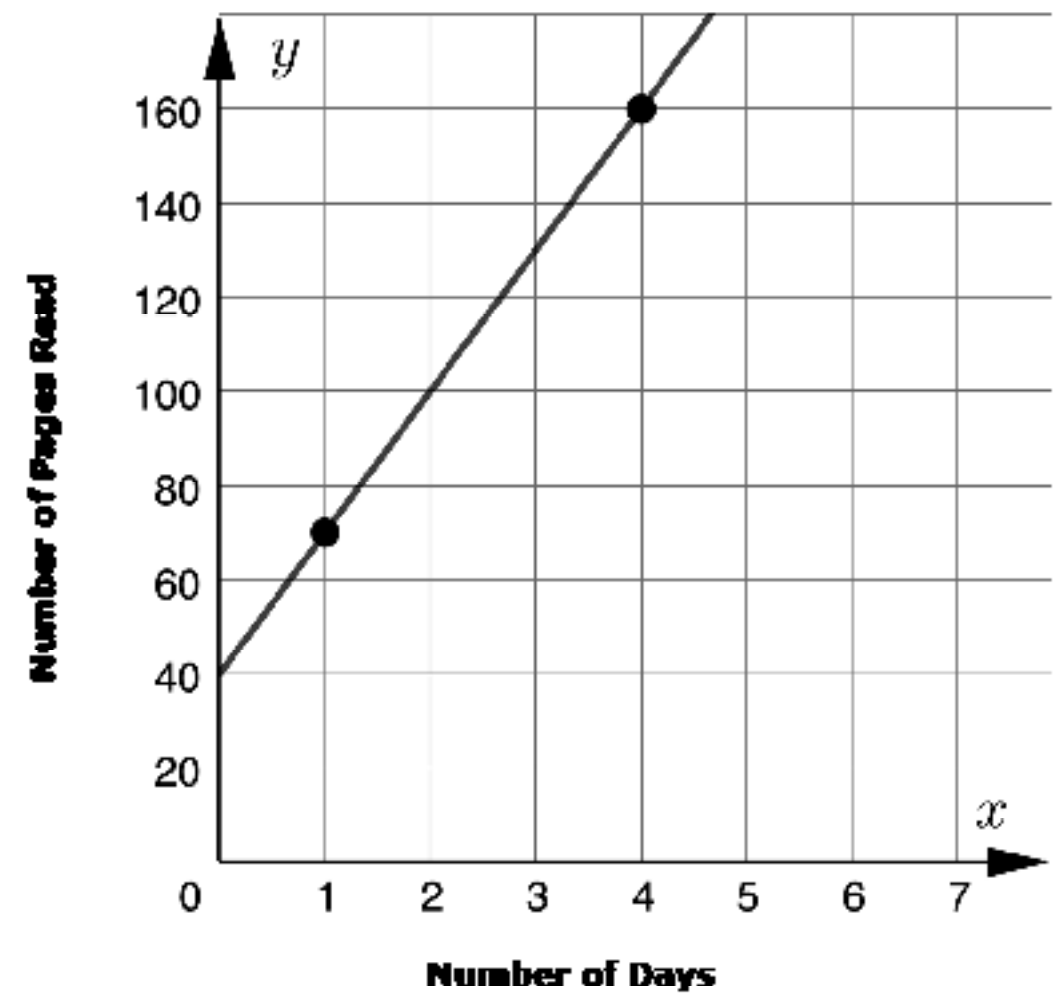
3. Customers at the local gym pay a \$59.99 membership fee to join and then a \$15.50 fee for each class they attend.

a. Complete the table with the total costs per classes attended.

Classes Attended	Total Cost
0	
1	
2	
3	
4	
5	
6	
7	

b. Use evidence from the written description or the table to explain if the situation is a proportional relationship or not.

4. Ashley is reading a book for book club. After reading the first 40 pages of the book in a week, she plans to read 30 pages each day until she finishes to complete the book in time for her next book club meeting. The graph of the situation is shown.



Ashley’s friend said the situation is proportional because she reads the same number of pages each day and the graph is a straight line. Use the graph to correct her friend.

6.3.4 Wrap-Up: Ticket Out the Door

1. Mark each relationship as either proportional or not proportional.

Relationship	Proportional or Not Proportional?								
$y = 1.7x$	☐ proportional ☐ not proportional								
	☐ proportional ☐ not proportional								
$y = \frac{2}{2}x$	☐ proportional ☐ not proportional								
<table><tr><th>x</th><th>y</th></tr><tr><td>3</td><td>1</td></tr><tr><td>6</td><td>2</td></tr><tr><td>9</td><td>3</td></tr></table>	x	y	3	1	6	2	9	3	☐ proportional ☐ not proportional
x	y								
3	1								
6	2								
9	3								
$y = 2x - 7$	☐ proportional ☐ not proportional								
	☐ proportional ☐ not proportional								

